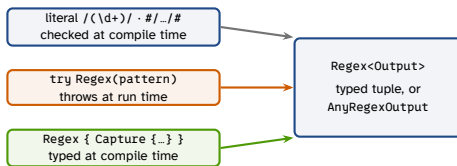


In one line: `Regex<Output>` (Swift 5.7, iOS 16): **Output** is the *typed* shape of a match, (Substring, captures...). Three builds: **literal** (checked + typed at compile time), `Regex(string)` (runtime, throws, `AnyRegexOutput`), **RegexBuilder** (typed, composable, embeds Foundation parsers). Matches **Characters**, returns `String.Index` ranges.

three ways in → one type



one pattern `/\d+/`, four questions, input `"ab12c345"`

<code>firstMatch(of:)</code>	a b 1 2 c 3 4 5	"12"
<code>matches(of:)</code>	a b 1 2 c 3 4 5	["12", "345"]
<code>wholeMatch(of:)</code>	a b 1 2 c 3 4 5	nil – must consume ALL
<code>prefixMatch(of:)</code>	a b 1 2 c 3 4 5	nil – must start at index 0

`contains(_:) · ranges(of:) · replacing(_:with:) · split(separator:) · trimmingPrefix(_:)` all take a regex

How it works

- Literal:** parsed by the compiler — bad pattern = compile error; captures become tuple elements, (`?<year>...`) a *labelled* one (`m.year`). Bare `/.../` needs Swift 6 mode or `BareSlashRegexLiterals` (SE-0354); `#/.../#` always works.
- Runtime** `try Regex(str)` for user/config patterns: `m.output[1].substring, m.output["y"]?.substring`.
- RegexBuilder** (import `RegexBuilder`): `OneOrMore`, `ZeroOrMore`, `Optionally`, `Repeat(2...4)`, `ChoiceOf`, `Anchor`, `Lookahead/NegativeLookahead`, `CharacterClass` (`.digit`, `.anyOf("+-")`). A literal can sit inside.
- Capture adds a tuple element (transform can map it); `TryCapture`'s transform returns `T?` — nil **fails that attempt and backtracks**: parse + validate in one step. `Reference(T.self)` names a capture: `m[ref]`.
- Foundation parsers are components** (SE-0357 `CustomConsumingRegexComponent`): `.date(.numeric, locale:, timeZone:)`, `.localizedCurrency(code:locale:) → Decimal`, `.iso8601(...)` — typed `Date/Decimal` captures, no second parse. Always pass locale + time zone.
- Options return a new regex: `.ignoresCase()`, `.dotMatchesNewlines()`; `.matchingSemantics` switches graphemes (default: `e+U+0301` is *one* .) to scalars. `.reluctant = lazy +?`; Local `{...}` = atomic group. The engine **backtracks**.

Example — typed literal + builder

```

let ym = /(??<y>\d{4})-(??<mo>\d{2})/ // (Substring, y:, mo:)
if let m = "on 2026-09".firstMatch(of: ym) { print(m.y, m.mo) }

import RegexBuilder
let cents = Reference(Int.self)
let line = Regex {
  Capture { /CREDIT|DEBIT/ } // Substring
  OneOrMore(.whitespace)
  Capture(.date(.numeric, locale: Locale(identifier: "en_US"),
    timeZone: .gmt)) // Date
  OneOrMore(.whitespace)
  TryCapture(as: cents) { OneOrMore(.digit) }
  transform: { Int($0) } // overflow: nil, no match
}
if let m = "DEBIT 3/5/2026 1250".wholeMatch(of: line) {
  let (_, kind, date, _) = m.output; print(kind, date, m[cents]) }
  
```

vs NSRegularExpression

	Swift Regex	NSRegularExpression
units	Character (graphemes)	UTF-16 code units (ICU)
ranges	<code>Range<String.Index></code>	<code>NSRange → Range(r, in: s)</code>
captures	typed tuple	<code>range(at:)</code> ; unmatched <code>NSNotFound</code>
OS	iOS 16+	all; runtime errors; \$1 templates

```

let re = try NSRegularExpression(pattern: #"(\d{4})"#)
let all = NSRange(s.startIndex..., in: s) // NOT s.count
if let m = re.firstMatch(in: s, range: all),
  let r = Range(m.range(at: 1), in: s) { print(s[r]) }
  
```

Performance & anchoring

- Build once (a stored `let`), never in a loop. `firstMatch` retries at every index — anchor (`^`, `Anchor.startOfSubject`) or use `whole/prefixMatch`.
- Catastrophic backtracking:** `/(a+)+$/` on `"aaaa...b"` is exponential — no nested quantifiers, Local. **No timeout API:** untrusted pattern = ReDoS.

When NOT to use a regex

Fixed text (`contains`, `hasPrefix`, `split`) · JSON (`Codable`) · URLs (`URLComponents`) · ISO dates (`ISO8601FormatStyle`) · nested/balanced input — regex can't count.

Traps

- ! `"abc123".wholeMatch(of: /\d+/)` is nil: whole = implicitly anchored both ends.
- ! `NSRange(location: 0, length: s.count)` — count is Characters, `NSRange` wants UTF-16: emoji break it. Use `NSRange(s.startIndex..., in: s)`.
- ! `Regex(str)` for a fixed pattern forfeits compile-time checks and typed captures. `/a*/` “finds” an empty range at 0.

Remember

“Literal = typed, String = throws, Builder = composes.”
Whole anchors, first scans.

Related: `strings-and-collections` (graphemes, `String.Index`) · `macros-and-result-builders` (`RegexBuilder` is a result builder) · `swift-numeric-overflow` (`Decimal`) · `FormatStyle` / `ParseStrategy`